

# AJAY PATEL

Software Engineer

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## Summary

Software Engineer with two years shipping **production LLM, RAG, and multi-agent systems** for a background-verification platform serving **50+ clients** - retrieval and LLM adjudication over 200M+ records, autonomous agents, and RAG chatbots on AWS Bedrock, MCP, and vector search (Qdrant, pgvector).

## Experience

### Equal Identity Private Limited

Hyderabad, India

Software Engineer

June 2024 - Present

- Built a **multi-layered LLM workflow** for an in-house legal/court-records engine: deterministic fuzzy retrieval narrows a **200M+ record** lake, an LLM (AWS Bedrock) adjudicates hard matches, and rules drive severity - cutting checks from days to **~15 minutes** at **~95% lower cost**, holding **~98% precision**.
- Built an autonomous **multi-agent** insufficiency-resolution system (AWS Bedrock, Agno) that detects and collects missing documents via email and MCP-based OCR (Textract) - cutting manual handling by **~60%**.
- Shipped an **autonomous AI voice agent** (Ultravox) that phones candidates in natural language to collect missing details end-to-end - no human operator in the loop.
- Created an internal **LLM chatbot** for Equal Console (AWS Bedrock) with tool-calling for verifications, notifications, and ticketing - grounded via **RAG (Qdrant)**, resolving **~30%** of HR/support queries end-to-end.
- Built an **MCP server for Equal Console** exposing verification, notification, and ticketing operations as LLM tools, and **LLM evaluation harnesses** (precision/recall, regression checks) that gate model and prompt changes to production.

## Projects

### In-House Legal / Court-Records Verification Engine

*Python, FastAPI, AWS (Bedrock, S3, Athena)*

- Crawls and indexes **200M+** Indian eCourts records into an S3 + Parquet data lake (Athena); **two-stage retrieve-and-rerank** (deterministic retrieval feeding LLM adjudication) took criminal-record checks fully in-house - multi-day vendor process to **~15 minutes** at **~95% lower cost**.

### ArchCritic - AI System Design Interview Coach ([live](#), [GitHub](#))

*Python, FastAPI, pgvector, Next.js*

- Live RAG product**: staged mock interviews graded per rubric dimension by an LLM judge grounded in real engineering blogs **with citations**; the architecture whiteboard is a structured graph the judge reads directly.
- Hybrid retrieval** (pgvector HNSW + FTS via RRF, cross-encoder rerank), local BGE embeddings; **eval-gated in CI** - retrieval golden set (**recall@5 0.87**) and judge calibration; live on Vercel + HF Spaces + Supabase.

### AI Mail Assistant ([live](#))

*Next.js, Gmail API, OAuth*

- Gmail-backed mail client where an LLM **controls the UI** via 9 tool-calls - fills compose forms, filters the inbox, drafts context-aware replies - with human-in-the-loop send confirmation.

## Technical Skills

- LLM & RAG**: LLMs, Retrieval-Augmented Generation, Embeddings, Semantic Search, **Qdrant** (vector database), Prompt Engineering, Evaluation
- Agentic AI**: Multi-Agent Systems, Tool-Calling, Model Context Protocol (MCP), AWS Bedrock, Agno, Ultravox (voice), Textract (OCR)
- Languages & Frameworks**: Python, Java, SQL, C++, FastAPI, Spring Boot, React/Next.js
- Cloud & Data**: AWS (Bedrock, Lambda, SQS, SNS, DynamoDB, S3, ECS, Athena, Textract, CDK), Data Lakes (S3 + Parquet), Docker, CI/CD
- Architecture**: Microservices, Event-Driven, REST APIs, gRPC

## Achievements

- Magichthon Hackathon ([GitHub](#))**: Built and shipped a solo **vision-LLM** meme-maker app in a day (Next.js, Supabase, OpenRouter) - **selected and advanced to the interview round**.

## Education

### Indian Institute of Technology Hyderabad (IITH)

Bachelor of Technology

July 2020 - July 2024

CGPA: 8.05 / 10